

Fengxiaoxiao Li

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RESEARCH INTERESTS

Data-driven and learning-based control • Machine learning in closed-loop systems • Safe and distributed multi-agent robotics • Event-triggered control & communication • Optimization and reinforcement learning for robotic decision-making

RESEARCH PROFILE

MSc student at NUS (CDE) in the Multi-Agent Robotic Motion (MARMot) Laboratory. I work on learning-driven multi-objective routing/task allocation and communication-efficient multi-agent formation control. Interested in uncertainty-aware learning and safety mechanisms for intelligent robotics.

EDUCATION

**National University of Singapore
(NUS), College of Design and
Engineering**

MSc in Mechanical Engineering; GPA:
4.63/5.00 (Sem 1)

Research: Multi-Agent Robotic
Motion (MARMot) Laboratory

Aug 2025 – Jul 2026

**University of Electronic Science
and Technology of China
(UESTC)**

BEng in Aircraft Control and
Information Engineering; GPA:
3.78/4.00 (Ranked 4/22)

Honor: Exempted from National
Postgraduate Entrance Examination
due to academic excellence

Aug 2021 – Jul 2025

RESEARCH EXPERIENCE

**Learning-Driven Optimization for
MOMTSP and MOHCVRP —**

NUS MARMot Lab

Aug 2025 – Present

Advisor: Asst. Prof. Guillaume Sartoretti (NUS)

Focus: scalable learning in closed-loop decision-making for multi-agent robotic routing and task allocation.

- Built a Transformer-based PARCO framework to generate diverse Pareto-optimal solution sets for multi-objective routing.
- Implemented multi-agent RL with sparse rewards and transformer-based inter-agent communication for cooperative decision-making.
- Evaluated against NSGA-III/MOEA baselines with Pareto-front analysis, showing strong hypervolume and fast inference.

**CAMO: Scalable Neural Solver
for Multi-Objective MTSP
(MOMTSP)** — NUS MARMot Lab

Aug 2025 – Nov 2025

Advisor: Asst. Prof. Guillaume Sartoretti (NUS)

Relevance: data-driven optimization under multi-objective trade-offs, emphasizing generalization and reliability.

- Achieved zero-shot generalization across arbitrary problem sizes via conditional attention and a conditional encoder.
- Proposed a decoupled dual decoder (agent selection / node selection) to remove input-dimension constraints and improve scalability.
- Enhanced REINFORCE with dynamic sampling; improved hypervolume and reduced inference time from hours to seconds.

**Prescribed-Time Formation
Tracking with Neural
Event-Triggered Control** —
UESTC

May 2024 – Jun 2025

Advisor: Asst. Prof. Zhiqiang Li (UESTC)

Relevance: communication-efficient distributed control with a pathway to safety- and uncertainty-aware robotics.

- Designed a prescribed-time, time-varying formation tracking protocol for second-order multi-agent systems with Lyapunov proofs.
- Developed a neural adaptive event-triggered mechanism to reduce communication while maintaining tracking accuracy.
- Validated robustness in MATLAB/Simulink (ode45) under communication noise and external disturbances.

**All-Terrain Foldable
Multi-Legged Exploration Robot**

— UESTC

Mar 2023 – Jul 2023

Advisor: Prof. Zhenwei Wang (UESTC)

Relevance: safe intelligent robotics via end-to-end integration of perception, control, and edge deployment.

- Designed a foldable hexapod and implemented a stable triangular gait for uneven terrain (SolidWorks/ProE).
- Built embedded locomotion control (Arduino UNO + TB6600) for NEMA 23 stepper motors with precise speed/steering.
- Trained CNN-based obstacle recognition (96.8% accuracy) and deployed optimized C++ inference on ESP32-CAM.

ADDITIONAL ML EXPERIENCE

Pneumonia detection from chest X-ray images (NTU summer program): trained a CNN in TensorFlow (95.63% accuracy) and benchmarked against ResNet/Random Forest/Logistic Regression.

PUBLICATIONS

- **Fengxiaoxiao Li***, Xiao Mao*, et al. “CAMO: A Scalable Neural Solver for the Multi-objective MTSP.” Submitted to the 15th International Conference on Swarm Intelligence (ANTS), 2026.

HONORS & AWARDS

- First Prize, National Graduation Design Competition for UAV Engineering (Jun 2025).
- Second Prize, National Undergraduate Thesis Competition (Aeronautics & Astronautics) (May 2025).
- UESTC Outstanding Student Scholarship (2022–2024) and Yunhui Specialized Scholarship (Top 5%) (Oct 2023).
- Distinction Winner, NTU AI Lab Summer Research Program (Top 2%) (Jul 2024).
- Top 15.6% globally, IEEE Xtreme 18.0 Programming Competition (907/5,800+ teams) (Oct 2024).
- Graduate Awards: Innovation/Endeavor/Contribution (Selected 1/23) (Jun 2025).

TECHNICAL SKILLS

Programming: Python, MATLAB, LaTeX, Markdown

Control & Learning: RL (DQN, PPO, SAC, DDPG, A3C, MADDPG), multi-agent systems, optimization, event-triggered control

Tools: MATLAB/Simul

ink, SolidWorks, Arduino, Altium, Keil uVision5, Git, Jupyter, PyCharm, VS Code, TensorFlow, PyTorch, Scikit-Learn, OpenCV

INTERESTS

Table tennis, football, swimming, running, reading (AI/robotics/sci-fi), soulslike games.